

SUMMARY

AI research and systems engineer with 6+ years of experience in multimodal pre-training and post-training, data curation, evaluation, open-source model integration, and production deployment. Built PyTorch training/serving systems and contributed 40+ pull requests to Hugging Face Transformers.

EXPERIENCE

- Toss Bank** Seoul, South Korea
 - Data Scientist* *Jan 2026 - Present*
 - On-premise AI Agent System:** Building an on-premise agent system with internally deployed LLMs for secure AI-assisted development workflows.
 - AI Authentication:** Developing AI-based authentication systems for face and ID card verification.
 - Document AI:** Post-trained and evaluated an approximately 1B-parameter vision-language model in an end-to-end document extraction pipeline; designed stage-level diagnostics and reached 61% exact-match accuracy for automation-ready outputs.
- SuperbAI** Seoul, South Korea
 - Machine Learning Engineer* *Sep 2021 - Jan 2026*
 - [Lead] Vision Foundation Model:** Led multimodal pre-training and staged text/image alignment for a visual-grounding model; curated 1.1M images with captions, boxes, and masks, then drove evaluation, API deployment, and AWS Marketplace release.
 - CVPR 2025 Competitions:** Won 2nd place in IOD and 4th place in FSOD challenges, validating the model on open evaluation benchmarks.
 - [Lead] ML Training & Serving Platform:** Built distributed, multi-GPU training infrastructure with AWS Batch and production serving with TensorRT/Triton; improved throughput by 5x and delivered 1,100+ models and 60+ customer endpoints in one year.
 - Post-training and Efficiency:** Applied LoRA, Adapter, and LST methods and memory-efficient training, reducing GPU memory by 65.6% and training time by 44.3% while preserving model performance.
 - Interactive Segmentation:** Built browser, hybrid, and cloud-based segmentation tools using RepViT-SAM, FocalClick, SAM, and SAM2, increasing segmentation labeling speed by 25x.
- Kakao Enterprise** Pangyo, South Korea
 - Machine Learning Engineer Intern* *Feb 2021 - Aug 2021*
 - AutoML:** Fixed AutoGluon NeuralNetFastAI scaling issue and developed a Flask-based training/inference AutoML framework with a simple front-end.
- Team ISLAND** Seoul, South Korea
 - Chief Technology Officer* *Mar 2019 - Oct 2020*
 - Startup:** Raised approximately \$290K through government subsidies and seed funding; managed five developers across Android, Unity, and deep learning workstreams.
 - Mobile ML:** Developed a lightweight 3D human pose estimation model for mobile devices and converted PyTorch models to TensorFlow Lite.

OPEN SOURCE

- Hugging Face Transformers:** Contributed 40+ pull requests to a large production codebase across multimodal and vision models, including model architectures, processors, conversion scripts, distributed training fixes, tests, and documentation.
- Hugging Face Transformers - SAM2:** Led the addition of Segment Anything 2 support to Transformers, including model integration, image/video processors, conversion flow, documentation, and tests; credited in the official [SAM2 docs](#).
- Hugging Face Transformers - Molmo2:** Opened Molmo2 support for Transformers and published [danelcsb/Molmo2-4B](#) on the Hugging Face Hub.

PAPERS

- **ZERO: Multi-modal Prompt-based Visual Grounding:** arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices:** IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array:** UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation:** Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease:** Sejong Science High School - Dec 31, 2013

EDUCATION

- **KAIST (Korea Advanced Institute of Science and Technology)** Daejeon, South Korea
Professional Master in Electrical Engineering; GPA: 3.55/4.3 Feb. 2020 – Aug. 2021
- **POSTECH (Pohang University of Science and Technology)** Pohang, South Korea
Bachelor in Electrical Engineering; GPA: 3.4/4.3 Mar. 2015 – Feb. 2020
- **UIUC (University of Illinois Urbana-Champaign)** Illinois, United States
Exchange Student; GPA: 3.6/4.0 Jan. 2018 – Dec. 2018

PROGRAMMING SKILLS

- **Research:** Python, PyTorch, Transformers, multimodal pre/post-training, LoRA, contrastive alignment
Systems: AWS Batch, multi-GPU training, MLflow, TensorRT, Triton, Git
Agents: LangChain, Langfuse, Claude Code, Codex, opencode **Languages:** Korean (native), English (professional)